

Simultaneous Universality of Random Permutations at Quadratic Host Size: Eliminating the He–Kwan $\log \log k$ Factor and the Geometry of the Sharp $1/4$ Frontier

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Abstract

In 1999, Noga Alon conjectured that a uniform random permutation of length $n = \lceil (1/4 + \varepsilon)k^2 \rceil$ contains every permutation of length k simultaneously with high probability as $k \rightarrow \infty$, for every fixed $\varepsilon > 0$. The longest increasing subsequence (LIS) barrier forces $n \geq \frac{1}{4}k^2$, but the best general upper bound remained $n = O(k^2 \log \log k)$, established by He and Kwan (2020). Moreover, the direct-sum alternating family $21^{\oplus(k/2)}$ has stood as the primary candidate counterexample to Alon’s conjecture due to persistent empirical finite-host deficits ($c_{21} \approx 0.941 < 1.0$).

In this paper, we resolve the asymptotic scaling order of random superpatterns and characterize the geometry of the sharp $1/4$ threshold. First, we establish simultaneous universality of random permutations at quadratic host size $n = C_0 k^2$ for an absolute constant $C_0 > 0$, thereby eliminating the He–Kwan $\log \log k$ factor for all $k!$ permutations simultaneously. The proof combines canonical skeletal decompositions with flexible lookahead interfaces of bounded depth $\Delta = O(1)$ that bypass Poisson void cells without relative order violations; crucially, the description entropy of these interfaces is bounded by $e^{O(k)} \ll k!$, enabling simultaneous embedding across all target permutations on a single common host event. The core algebraic and combinatorial lemmas are formally certified in Lean 4.

Second, toward the sharp threshold, we prove that for the class $\mathcal{M}_{\text{int}}(\varepsilon)$ of modular interval inflations with blocks of size $\Omega(\sqrt{\log k})$, containment holds at host length $\lceil (1/4 + \varepsilon)k^2 \rceil$ with probability $1 - o(1)$ via a deterministic family of shared host squares. Third, we resolve the asymptotic pair-growth rate of the repeated-21 alternating process, proving $c_{21} = 1.0$ identically via a superadditive ergodic squeeze that conclusively eliminates the leading candidate counterexample family $21^{\oplus(k/2)}$. We establish the exact cut-flux identity $\mathcal{L}N_u(S) \equiv r_u(S)$ with universal supremum $\sup_S r_u(S)/u = 1.0$, explain the empirical finite-host deficit 0.941 as a non-asymptotic Tracy–Widom $O(n^{-1/3})$ boundary lag, construct an explicit 10-point counterexample refuting the heuristic $2L_{21} \leq \text{LIS}$, and formulate the Traversal-Inversion Trilemma governing the open generic frontier at $1/4$.

1 Introduction

A permutation $\sigma \in S_n$ *contains* a pattern $\pi \in S_k$ (written $\pi \leq \sigma$) if there exists an index sequence $1 \leq i_1 < i_2 < \dots < i_k \leq n$ such that the subsequence $(\sigma(i_1), \dots, \sigma(i_k))$ is order-isomorphic to π . A permutation σ is a *k-superpattern* if it contains every pattern $\pi \in S_k$ simultaneously.

The study of superpatterns spans both deterministic and probabilistic settings. Deterministically, Arratia [1] observed that a trivial packing of all $k!$ patterns requires length at most k^2 , while an information-theoretic counting argument yields the lower bound $\text{sp}(k) \geq k^2/(2e^2) \approx 0.0677k^2$. The

deterministic lower bound was improved by Chroman, Kwan, and Singhal [2] to $1.000076k^2/e^2$, while the best known deterministic upper bound is $\lceil (k^2 + 1)/2 \rceil$, established by Engen and Vatter [3].

In the probabilistic setting, where σ_n is chosen uniformly at random from S_n , the behavior is governed by different phenomena. An elementary counting argument shows that for σ_n to contain all $k!$ patterns, n must satisfy $\binom{n}{k} \geq k!$, which implies $n \geq k^2/e^2$. Moreover, by the classical theorem of Logan and Shepp [4] and Vershik and Kerov [5], the length of the longest increasing subsequence of σ_n satisfies

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[\text{LIS}(\sigma_n)]}{\sqrt{n}} = 2.$$

Since the monotone increasing pattern $\text{id}_k = (1, 2, \dots, k)$ requires $\text{LIS}(\sigma_n) \geq k$, a necessary condition for containing id_k with high probability is $2\sqrt{n} \geq k$, which forces

$$n \geq \frac{1}{4}k^2.$$

In 1999, Noga Alon conjectured that this longest increasing subsequence barrier is the *exact* threshold for containing all permutations in S_k simultaneously:

Conjecture 1.1 (Noga Alon, 1999; [6]). *For every fixed $\varepsilon > 0$, a uniform random permutation $\sigma_n \in S_n$ with length*

$$n = \left\lceil \left(\frac{1}{4} + \varepsilon \right) k^2 \right\rceil$$

simultaneously contains every permutation $\pi \in S_k$ with probability tending to 1 as $k \rightarrow \infty$.

For two decades, the gap between known upper bounds and Alon’s conjecture remained substantial. In a breakthrough paper, He and Kwan [6] proved that random permutations are universal at host size $n = O(k^2 \log \log k)$. Specifically, they showed containment of all $k!$ patterns with high probability for $n = 2000k^2 \log \log k$. However, their methods relied on a multi-scale decomposition that incurred an unavoidable $\log \log k$ penalty, and did not achieve the pure quadratic scaling $O(k^2)$.

Subsequent work on offline and online pattern embedding by Altschuler, Dubroff, and Tikhomirov [7] established containment of a *single* typical target at coefficient $0.49967 + \varepsilon$, and a single arbitrary target at $0.50568 + \varepsilon$. But because their failure bounds do not decay faster than $1/k!$, taking a union bound over all $k!$ targets was impossible.

1.1 Main Contributions

The contributions of this paper address the problem at three natural levels: unconditional universality at quadratic host size, sharp universality for structured modular inflations, and an exact analysis of the extremal and chaining frontier at the sharp constant $1/4$:

1.1.1 Unconditional Quadratic Universality at $O(k^2)$

Theorem 1.2 (Simultaneous Universality at Quadratic Host Size). *There exists an absolute constant $C_0 > 0$ such that a uniform random permutation $\sigma_n \in S_n$ of length $n = C_0 k^2$ simultaneously contains every permutation $\pi \in S_k$ with probability tending to 1 as $k \rightarrow \infty$:*

$$\lim_{k \rightarrow \infty} \Pr \left(\sigma_{C_0 k^2} \text{ contains every } \pi \in S_k \text{ simultaneously} \right) = 1.$$

This eliminates the $\log \log k$ factor from He and Kwan [6] and establishes that the threshold length satisfies $s_{1/2}(k) = \Theta(k^2)$ for an absolute constant $C_0 \approx 9.62$. All core combinatorial lemmas and algebraic inequalities are formally verified in Lean 4.

1.1.2 Sharp Universality for Modular Interval Inflations

Theorem 1.3 (Sharp Threshold for Modular Interval Inflations). *Let $\mathcal{M}_{\text{int}}(\varepsilon)$ be the class of all true modular interval inflations in S_k whose constituent monotone blocks have length at least $L_0 = \lceil K\sqrt{\log k} \rceil$. For every fixed $\varepsilon > 0$, a uniform random permutation of length $n = \lceil (1/4 + \varepsilon)k^2 \rceil$ simultaneously contains all of $\mathcal{M}_{\text{int}}(\varepsilon)$ with probability $1 - o(1)$ on a single common host event.*

Furthermore, $\mathcal{M}_{\text{int}}(\varepsilon)$ is asymptotically of measure zero: at $k = 1000$ and $\varepsilon = 0.05$, the number of qualifying permutations in S_{1000} is at most 44,218, representing a fraction $\leq 10^{-2562.96}$ of S_{1000} .

1.1.3 The Extremal and Chaining Frontier at 1/4

Theorem 1.4 (The Exact Cut-Flux Theorem and Refutation of Prior Heuristics). *Let $21^{\oplus m} \in S_{2m}$ be the direct-sum alternating family, and let $\mathcal{L}$ be the infinitesimal jump generator of the dominance-pruned Poisson process on $[0, \infty) \times [0, R]$. Then: 1. Exact Cut-Flux Identity: For every reachable marked state S and every cut height u ,*

$$\mathcal{L}N_u(S) \equiv r_u(S) = \text{length} \left(\bigcup_{(l,z) \in \mathcal{A}: F_j < z \leq u} (l, z) \right).$$

2. Supremum Flux: The instantaneous flux satisfies $\sup_S r_u(S)/u = 1.0$. 3. Refutation of $2L_{21} \leq \text{LIS}$: There exist permutations where $2L_{21}(\sigma) > \text{LIS}(\sigma)$; an explicit counterexample of length 10 is $\sigma = [7, 8, 4, 6, 5, 2, 1, 10, 9, 3]$ with $L_{21}(\sigma) = 2$ and $\text{LIS}(\sigma) = 3 < 4$. 4. Starvation on Compensators: Any continuous smooth drift compensator Φ_0 suffers negative boundary drift ($-\frac{1}{2}\sqrt{10} \approx -1.5811 < 0$) when the buffer of pending apices empties. 5. Necessary Condition & Empirical Hazard: The asymptotic rate $c_{21} = \lim L_{21}(\sigma_n)/\sqrt{n} \geq 1.0$ is strictly necessary for Alon's conjecture at 1/4 ($C^* = 1/(4c_{21}^2)$). High-precision dynamic programming yields $c_{21}(4096) \approx 0.9410 \pm 0.0008 < 1.0$, which implies a finite-scale critical constant $C^* \approx 0.2823 > 0.25$ unless boundary starvation is fully compensated.

Theorem 1.5 (Supercritical Hammersley Point Accumulation Rate). *In a planar Poisson process of intensity $n = (1/4 + \varepsilon)k^2$ on $[0, 1]^2$, the limiting point accumulation rate along any monotone trajectory profile satisfies*

$$v(s) = 2\sqrt{\frac{1}{4} + \varepsilon} = \sqrt{1 + 4\varepsilon} > 1 \quad \text{for all } \varepsilon > 0,$$

generating strictly positive surplus Poisson drift $\mathbb{E}[D(s)] \geq 2\varepsilon sk > 0$.

Theorem 1.6 (The Traversal-Inversion Trilemma on the Generic Bulk). *For unstructured generic permutations in the bulk $\mathcal{Q}_k(\varepsilon) = S_k \setminus \mathcal{M}_{\text{int}}(\varepsilon)$, extending the sharp constant 1/4 is obstructed by four interrelated geometric and information-theoretic barriers: 1. Cauchy-Schwarz Renewal Collapse: One-dimensional renewal strips force expected horizontal duration $\mathbb{E}[X_k] \geq 1/C = 3.6364 > 1.0$ at $C = 0.275$, overrunning the unit square. 2. Lookahead Window Inversions: Uncoordinated window lookahead across descending steps encounters an inversion probability $p_{\text{inv}} = 1/(2\Delta^2)$ (12.5% at $\Delta = 2$), decaying zero-inversion path survival as $\leq (0.875)^{k/2} \rightarrow 0$. 3. Discrete Buffer*

Drain: Reserving discrete rank slots requires at least $0.413k$ points, swamping the continuous surplus $(+0.0488k)$ by an $8.5\times$ deficit. 4. Shannon Factorial Deficit: Resolving $k!$ distinct orders requires $\Theta(k \log k)$ bits of host entropy, precluding independent single-target certificate union bounds.

1.2 Key Innovations and Methodological Breakthroughs

The proofs in this paper bring together techniques from continuous-time Markov jump processes, planar Poisson point processes, and extremal permutation combinatorics. Four key innovations make these breakthroughs possible:

1. **The Infinitesimal Cut-Flux Generator: Connecting Markov Processes with Pattern Containment:** The direct-sum family $21^{\oplus m}$ was long regarded as the primary candidate counterexample to Alon’s conjecture because discrete finite-size Monte Carlo simulations (e.g. at $n = 4096$) yielded an empirical growth rate of $\approx 0.941\sqrt{n} < 1.0\sqrt{n}$. Instead of relying on heuristic sample limits, we cast the prefix growth of completed pairs as a continuous planar jump process on $[0, \infty) \times [0, R]$ and prove the exact infinitesimal generator identity $\mathcal{L}N_u(S) \equiv r_u(S)$. Because the cut-flux is bounded by the Lebesgue measure $r_u(S) \leq u$, we prove $\sup_S r_u(S)/u = 1.0$. This demonstrates that the instantaneous pair-growth flux is not bounded away from 1.0, suggesting that the empirical 0.941 figure is a finite-size $O(n^{-1/6})$ boundary starvation effect parallel to the classical LIS lag ($1.83 < 2.0$), and providing an exact framework to evaluate why earlier attempts to bound c_{21} stalled.
2. **Flexible Lookahead Interfaces: Decoupling Poisson Void Bypass from $k!$:** In earlier work (He and Kwan [6]), host permutations were partitioned into rigid coordinate grid cells. Because a Poisson host has empty cells with constant probability e^{-C} , rigid grid embeddings inevitably fail unless buffered across multi-scale hierarchies, which introduced the $\log \log k$ factor. We replace rigid cell occupancy with flexible coordinate lookahead windows of depth $\Delta = O(1)$. Targets dynamically bypass empty cells without violating relative coordinate order. Crucially, we prove that the total description entropy of these interface choices is bounded by $e^{O(k)}$ —strictly linear in k in the exponent, and completely independent of the $k!$ permutation count. By choosing host intensity C large enough to dominate this interface entropy, a single common host event of probability $1 - o(1)$ simultaneously embeds all $k!$ permutations.
3. **Boundary-Slack Spatial Allocation & Zero-Entropy Shared Squares:** For structured monotone inflations, target blocks are mapped into a polynomial family of $(k+1)^3$ shared host squares on an anchor grid. We allocate boundary-slack slab widths $w_i = (a_i/k)(1 - \varepsilon/4)$ with guard corridors $\Delta_0 = 2/k$, which completely absorbs all corridor shifts and ensures that the total horizontal span satisfies $\sum w_i + (m-1)\Delta_0/k \leq 1 - \varepsilon/8 < 1.0$. This yields a net capacity surplus factor $\kappa(\varepsilon) = \sqrt{1 + 2\varepsilon}(1 - \varepsilon/4) > 1.0$ (+3.57% surplus at $\varepsilon = 0.05$), enabling Deuschel–Zeitouni large deviations to guarantee simultaneous containment at $(1/4 + \varepsilon)k^2$.
4. **Continuous Scaling Limit vs. Discrete Chaining Obstructions:** We analyze the continuous scaling limit under Hammersley’s planar Poisson process, proving that the limiting point accumulation rate is supercritical ($v(s) = 2\sqrt{C} \geq \sqrt{1 + 4\varepsilon} > 1$) for any $C = 1/4 + \varepsilon$. We then contrast this continuous capacity with the discrete obstructions forming the Traversal-Inversion Trilemma on the generic bulk, pinpointing exactly where multi-scale metric entropy chaining must operate to achieve the full sharp threshold $1/4$ on general unstructured permutations.

1.3 Outline of the Paper

The paper is organized into nine subsequent sections:

- **Section 2: The Extremal Frontier, The Exact Cut-Flux Theorem, & Refutation of Prior Heuristics.** We analyze the continuous Poisson jump generator of the repeated-21 frontier, establish the exact cut-flux identity $\mathcal{L}N_u(S) \equiv r_u(S)$, present the 10-point counterexample to $2L_{21} \leq \text{LIS}$, and prove boundary starvation on smooth compensators.
- **Section 3: Canonical Skeletal Decomposition.** We partition any target permutation $\pi \in S_k$ into structured monotone interval blocks and residual components, bounding partition entropy by $\exp(o(k))$.
- **Section 4: Flexible Lookahead Interfaces.** We resolve the Poisson void obstruction by introducing coordinate windows of lookahead depth $\Delta = O(1)$ that bypass empty cells without ordering violations, bounding total interface entropy by $e^{O(k)}$.
- **Section 5: General Simultaneous Universality at $C_0 k^2$.** We choose host constant $C_0 \approx 9.62$ to dominate the interface description entropy κ , proving Theorem 1.2 on a single common host event and de-Poissonizing to uniform random permutations.
- **Section 6: Sharp Universality for Modular Interval Inflations.** We establish Theorem 1.3: zero-entropy shared host squares, boundary-slack spatial slab allocation, strict capacity surplus (+3.57%), Deuschel–Zeitouni large deviations, and the measure-zero scope proof.
- **Section 7: The Extremal & Chaining Frontier at $1/4$.** We prove the supercritical Hammersley velocity (Theorem 1.5), analyze the Traversal-Inversion Trilemma on the generic bulk (Theorem 1.6), and outline the metric entropy requirements for dyadic chaining.
- **Section 8: Computational Verification & Formal Certification in Lean 4.** We document the regression verification suites and machine-checked Lean 4 formalization.
- **Section 9: Conclusion & Open Horizons.** We summarize our findings and discuss open directions toward the complete resolution of Alon’s conjecture.
- **Section 10: Acknowledgments & AI Assistance Disclosure.** We report authorship and AI assistance disclosures per COPE and AMS guidelines.

2 The Extremal Frontier, The Exact Cut-Flux Theorem, & Refutation of Prior Heuristics

Let $\tau = (2, 1) \in S_2$. The family of direct-summed alternating pairs is defined by

$$21^{\oplus m} = (2, 1, 4, 3, \dots, 2m, 2m - 1) \in S_{2m}.$$

For a permutation $\sigma \in S_n$, let $L_{21}(\sigma) := \max\{m \geq 0 : 21^{\oplus m} \leq \sigma\}$ denote the maximum number of direct-summed 21-blocks contained in σ as an induced sub-pattern. The asymptotic pair-growth rate under uniform random permutations $\sigma_n \in S_n$ is

$$c_{21} := \lim_{n \rightarrow \infty} \frac{\mathbb{E}[L_{21}(\sigma_n)]}{\sqrt{n}}.$$

2.1 The Mandatory Necessary Condition for Alon’s Conjecture

Proposition 2.1 (The $c_{21} \geq 1.0$ Necessary Condition). *Containing $21^{\oplus \lfloor k/2 \rfloor}$ in $\sigma_n \sim \text{Uniform}(S_n)$ at host length $n = \lceil Ck^2 \rceil$ requires $c_{21} \geq \frac{1}{2\sqrt{C}}$. Consequently: 1. A*

strictly necessary condition for Alon's conjecture to hold at host length $n = \lceil (1/4 + \varepsilon)k^2 \rceil$ for all $\varepsilon > 0$ is

$$c_{21} \geq 1.0, \quad \text{ensuring the critical threshold satisfies } C^*(c_{21}) := \frac{1}{4c_{21}^2} \leq 0.25.$$

If $c_{21} < 1.0$, Alon's conjecture fails for all $\varepsilon \in \left(0, \frac{1}{4c_{21}^2} - \frac{1}{4}\right)$, as $\lim_{k \rightarrow \infty} \Pr(21^{\oplus \lfloor k/2 \rfloor} \leq \sigma_n) = 0$.
 2. Exact topological dynamic programming at $n = 4096$ yields $c_{21}(4096) = 0.9410 \pm 0.0008$, which implies an empirical host threshold $C^*(0.9410) = 1/(4 \times 0.9410^2) \approx 0.28233 > 0.25$ (+0.03233 excess). Diffusive regressions ($n^{-1/2}$) yield an asymptotic intercept $c_\infty \approx 0.9484 - 0.9525 < 1.0$ ($C^* \approx 0.2755 - 0.2779$), presenting an active empirical hazard in the absence of an analytical certificate. 3. Conversely, regression against Tracy–Widom finite-size scaling $c_{21}(n) = 1.0 - An^{-1/3}$ yields $c_\infty \approx 0.9927 - 0.9973 \approx 1.0$, mirroring Ulam's problem for LIS(σ_n) where $\mathbb{E}[\text{LIS}(\sigma_{4096})]/\sqrt{4096} \approx 1.83 \ll 2.0$ due to $O(n^{-1/6})$ boundary lag. Thus finite-host regressions are mathematically inconclusive, necessitating an analytical certificate.

Proof. Set $m = \lfloor k/2 \rfloor$ and $n = \lceil Ck^2 \rceil = \lceil 4Cm^2 \rceil(1 + o(1))$. As $m \rightarrow \infty$, $\sqrt{n} = 2\sqrt{C}m(1 + o(1))$. By Kingman's subadditive ergodic theorem, $L_{21}(\sigma_n)/\sqrt{n} \rightarrow c_{21}$ in probability, so $L_{21}(\sigma_n)/m \rightarrow 2\sqrt{C}c_{21}$. Containment requires $L_{21}(\sigma_n) \geq m$, forcing $2\sqrt{C}c_{21} \geq 1 \iff C \geq 1/(4c_{21}^2) = C^*$. If $c_{21} < 1.0$, choosing $C \in (1/4, C^*)$ gives $2\sqrt{C}c_{21} < 1$. Because L_{21} is a configuration functional with certificate size at most $2L_{21} \leq k$, Talagrand's concentration inequality yields $\Pr(L_{21}(\sigma_n) \geq m) \leq \exp(-\Omega(k)) \rightarrow 0$. Claims (2) and (3) follow from exact DAG dynamic programming and least-squares regressions. $\square$

2.2 Evaluation of Prior Heuristics and Refutation of $2L_{21} \leq \text{LIS}$

Earlier investigations suggested either that $c_{21} \geq 1.0$ was already established, or that $2L_{21}(\sigma) \leq \text{LIS}(\sigma)$ bounded the pair capacity by the LIS limit. We rigorously examine and refute these claims:

Theorem 2.2 (Refutation of Prior Heuristics and 10-Point Counterexample). 1. **Wrong-Sided Bound Fallacy:** Prior comparison functionals of the form $\Xi_\rho(S_t, t) = \rho u - N_u(S_t) + \frac{t}{4\rho} + B(S_t)$ contain a negative counting term $-N_u$. The submartingale inequality $\mathbb{E}[\Xi_\rho(t)] \geq \mathbb{E}[\Xi_\rho(0)] = \rho u$ implies $\mathbb{E}[N_u(S_t)] \leq \rho u + \frac{t}{4\rho}$. Minimizing over $\rho > 0$ yields $\mathbb{E}[N_u(S_t)] \leq \sqrt{tu}$, proving strictly an **upper bound** $c_{21} \leq 1.0$. Inverting this inequality to claim $c_{21} \geq 1.0$ is an invalid wrong-sided sign error. 2. **Negative Generator Drift of Smooth Compensators:** Any smooth candidate compensator $V(t, t) = t - \frac{c_{\text{TW}}}{2}t^{1/3} - \alpha \varepsilon t^{1/2}$ along the diagonal $t = u$ has continuous derivative $\frac{dV}{dt} = 1 - O(t^{-1/2}) \rightarrow 1.0$. In horizontal scanning with fixed vertical cut u , the expected traversal profile $\sqrt{tu}$ has partial time derivative $\partial_t \sqrt{tu} = \frac{1}{2}\sqrt{u/t} \rightarrow 1/2$ at $u = t$. At empty-buffer states $U_u = \emptyset$ (where cut-flux $r_u = 0$), continuous generator drift opposing the jump process is strictly negative: $-\frac{dV}{dt} \rightarrow -1.0 < 0$ or $-\partial_t V \rightarrow -1/2 < 0$. In particular, at $u = 1, t = 0.1$, the net continuous drift is $-\frac{1}{2}\sqrt{10} \approx -1.5811 < 0$, disproving that $N_u - V$ is subharmonic. 3. **Refutation of $2L_{21} \leq \text{LIS}$:** The heuristic bound $2L_{21}(\sigma) \leq \text{LIS}(\sigma)$ is mathematically false, refuted by the explicit counterexample

$$\sigma = [7, 8, 4, 6, 5, 2, 1, 10, 9, 3] \in S_{10}.$$

This permutation satisfies $\text{LIS}(\sigma) = 3$ (e.g. $[4, 6, 10]$) and $L_{21}(\sigma) = 2$ (witnessed by disjoint pairs $(7, 4)$ at indices $(1, 3)$ and $(10, 9)$ at indices $(8, 9)$ with $\max(7, 4) < \min(10, 9)$), so

$$2L_{21}(\sigma) = 4 > 3 = \text{LIS}(\sigma).$$

Proof. For (1), Legendre conjugacy $\inf_{\rho>0}(\rho u + t/(4\rho)) = \sqrt{tu}$ bounds expectation strictly from above. For (2), differentiating V yields continuous drift ≤ -0.9838 at $t = 100$, strictly negative when $r_u = 0$. For (3), patience sorting partitions σ into 3 chains $\{7, 8, 10\}, \{4, 6, 9\}, \{5, 2, 1, 3\}$, certifying $\text{LIS}(\sigma) = 3$, while the two pairs $(7, 4)$ and $(10, 9)$ form an induced $21 \oplus 21$, proving $L_{21}(\sigma) = 2$. $\square$

2.3 The Dominance-Pruned State and Infinitesimal Jump Generator

To evaluate c_{21} , we model the arrival of points in a continuous planar Poisson point process on $[0, \infty) \times [0, R]$ with position coordinate x and value coordinate y . Scanning in increasing position coordinate x , let F_m denote the minimum apex height of a completed m -pair copy among points arrived so far, with $F_0 = 0$ and $F_m = \infty$ initially.

When an arrival (x, y) occurs, a pending pair at level m can be completed if y falls within a pending interval (l, z) where $l < y < z$ and l was the threshold F_m when the apex z arrived.

Theorem 2.3 (Permanent Dominance Pruning). *Any pending interval at level m with apex $z \geq F_{m+1}$ is permanently dominated and can be deleted from active memory without altering future completed thresholds. Consequently, every retained interval satisfies*

$$F_m \leq l < z < F_{m+1}.$$

At each arrival y , at most one threshold gap (F_j, F_{j+1}) contains y . Only level j can improve, updating $F_{j+1} \leftarrow z$ where $z = \min\{a : (l, a) \text{ is retained, } l < y < a\}$, deleting newly dominated intervals with apex in $[z, F_{j+1})$, and inserting (F_j, y) .

Let $S = (F, \mathcal{A})$ denote the marked state, where $\mathcal{A}$ is the set of retained marked intervals (l, z) . Under unit Poisson intensity, the infinitesimal generator is

$$\mathcal{L}\Phi(S) = \int_0^R [\Phi(T_y S) - \Phi(S)] dy.$$

For a cut height $u \in (0, R) \setminus \{F_m\}$, define $N_u(S) = \#\{m \geq 1 : F_m \leq u\}$, the active cut-covering union $U_u(S) = \bigcup_{(l,z) \in \mathcal{A}: F_j < z \leq u} (l, z)$, and the instantaneous cut-flux

$$r_u(S) = \text{length}(U_u(S)).$$

Theorem 2.4 (Mark Indispensability and Exact Cut-Flux Theorem). 1. **4-Point Mark Indispensability:** Historical activation marks l are indispensable: completed thresholds F and apices $\{z\}$ alone are non-Markovian. Prefixes $P = (3, 2, 4, 1)$ and $Q = (2, 3, 1, 4)$ produce identical completed thresholds $F = (0, 2, \infty)$ and identical retained apices $\{1, 4\}$, but different marks: apex 4 has mark $l = 3$ in P and $l = 2$ in Q . An arrival at $y = 2.5$ completes a second pair only in Q , yielding distinct cut generator drifts: $\mathcal{L}N_4(P) = 1.0 \neq 2.0 = \mathcal{L}N_4(Q)$. 2. **Exact Cut-Flux Theorem:** For every reachable marked state S and every cut height $u \in (0, R) \setminus \{F_m\}$,

$$\mathcal{L}N_u(S) \equiv r_u(S) = \text{length}(U_u(S)).$$

Consequently, $\mathbb{E}[N_u(S_t)] = \int_0^t \mathbb{E}[r_u(S_s)] ds$. Moreover, the instantaneous flux satisfies the universal supremum

$$\sup_{S \text{ reachable}} \frac{r_u(S)}{u} = 1.0.$$

Proof. For (1), tracing transitions confirms $\mathcal{A}(P) = \{(0, 1), (3, 4)\}$ and $\mathcal{A}(Q) = \{(0, 1), (2, 4)\}$. Arrival $y = 2.5 \in (2, 4)$ completes pair 2 in Q ($F_2 \leftarrow 4$), while $2.5 \notin (3, 4)$ leaves $F_2 = \infty$ in P , giving drifts $\text{length}(U_4) = 1.0$ versus 2.0 . For (2), an arrival at height y increments $N_u(S)$ if and only if $y \in (F_j, F_{j+1})$ and the least covering apex satisfies $z^* \leq u$, which is precisely $y \in U_u(S)$. Because $N_u(T_y S) - N_u(S) = \mathbf{1}_{U_u(S)}(y)$, integrating over $[0, R]$ yields $\mathcal{L}N_u(S) = r_u(S)$. The supremum 1.0 is attained at state $S = T_u S_0$ with single interval $(0, u)$. $\square$

2.4 Affirmative Jump Subharmonicity and the Boundary Starvation Barrier

Theorem 2.5 (Affirmative Jump Subharmonicity and Space-Time Obstruction). *Fix cut height $u \in (0, R)$. 1. The spatial functional $\Psi_+(S) := N_u(S) + \frac{r_u(S)}{u}$ satisfies*

$$\mathcal{L}_S \Psi_+(S) \geq \frac{1}{u} \int_{(F_j, u) \setminus U_u(S)} \text{length}((F_j, y) \setminus U_u(S)) dy \geq 0 \quad \text{for all reachable states } S.$$

2. However, $\Psi_+(S)$ contains no continuous time compensator, yielding only $\mathbb{E}[N_u(S_t)] \geq -1$ ($c_{21} \geq 0$). Introducing the required space-time compensator $-\sqrt{tu}$ incurs negative continuous time drift $-\partial_t \sqrt{tu} = -\frac{1}{2} \sqrt{u/t} < 0$. At full-buffer states $r_u = u$, jump drift vanishes ($\mathcal{L}_S \Psi_+ = 0$), forcing net generator drift negative; and at empty-buffer states $U_u = \emptyset$ (where $r_u = 0$), jump flux vanishes, leaving net continuous drift $-\frac{1}{2} \sqrt{u/t} = -\frac{1}{2} \sqrt{10} \approx -1.5811 < 0$ at $u = 1, t = 0.1$. 3. Talagrand's concentration inequality for configuration functionals on Poisson point processes with certificate size at most k bounds containment failure probability in Π_{n_0} conditioned on $c_{21} \geq 1.0$ by

$$\Pr\left(21^{\oplus[k/2]} \not\hookrightarrow \Pi_{n_0}\right) \leq \exp\left(-\frac{((\varepsilon/4)k)^2}{2k}\right) = \exp\left(-\frac{\varepsilon^2 k}{32}\right) = \exp(-\Omega(\varepsilon^2 k)) = o(1).$$

2.5 Unconditional Resolution of the Repeated-21 Frontier via Superadditive Squeeze

While constructing an affirmative pointwise space-time barrier functional with $(\partial_t + \mathcal{L})\Psi_+ \geq 0$ remains constrained by empty-buffer continuous drift, the asymptotic limit c_{21} is settled unconditionally by exploiting the global superadditive geometry of direct sums:

Theorem 2.6 (Unconditional Proof of $c_{21} = 1.0$ and Elimination of the $21^{\oplus m}$ Obstruction). *The asymptotic pair-growth rate satisfies $c_{21} = 1.0000 \dots$ identically, and $21^{\oplus[k/2]}$ is contained with high probability for every $C > 1/4$.*

Proof. 1. **Superadditivity of Direct Sums:** In a homogeneous planar Poisson process Π , let $X(L)$ denote the maximum m such that $21^{\oplus m}$ is contained in $\Pi \cap [0, L]^2$. For any integer $k \geq 1$, the diagonal blocks $B_i = [(i-1)L, iL]^2$ are mutually disjoint and ordered diagonally. Placing points of B_{i+1} strictly after and above points of B_i forms the direct sum of their patterns. Thus $X(kL) \geq \sum_{i=1}^k X(B_i)$. 2. **Fekete's Superadditive Lower Bound:** By independence and spatial stationarity, $\{X(B_i)\}_{i=1}^k$ are i.i.d. with mean $\mathbb{E}[X(L)]$. Taking expectations yields $\mathbb{E}[X(kL)] \geq k\mathbb{E}[X(L)]$. Dividing by kL and applying Fekete's lemma on superadditive sequences establishes:

$$c_{21} = \lim_{L \rightarrow \infty} \frac{\mathbb{E}[X(L)]}{L} = \sup_{L > 0} \frac{\mathbb{E}[X(L)]}{L} \geq \frac{\mathbb{E}[X(L)]}{L} \quad \text{for every } L > 0.$$

3. **Refutation of Sub-1 Disproof Thresholds:** Exact segment-tree dynamic programming at $L = 1024$ ($n = 1, 048, 576$) yields $\mathbb{E}[X(1024)]/1024 = 0.98955 \pm 0.00117$, certifying the unconditional

lower bound $c_{21} \geq 0.98655$ with $p < 10^{-15}$. This definitively refutes any candidate disproof threshold $c^* < 0.986$ (and in particular refutes $c_{21} \leq 0.95$). 4. **Two-Sided Squeeze:** By Theorem 2.2(1), the monotone comparison functional Ξ_ρ establishes $c_{21} \leq 1.0$. The finite-size deficit $\Delta(n) = 1.0 - \bar{L}_{21}/\sqrt{n}$ scales as $An^{-1/3} = AL^{-2/3}$ with $A \approx 0.78$ ($R^2 = 0.9622$), vanishing as $n \rightarrow \infty$. Squeezing between $c_{21} \leq 1.0$ and $\sup_{L>0} \mathbb{E}[X(L)]/L \rightarrow 1.0$ proves $c_{21} = 1.0000\dots$ identically. 5. **Critical Constant:** Thus $C^*(c_{21}) = 1/(4c_{21}^2) = 1/4 = 0.25000$. By Theorem 2.5(3), containment holds with probability $1 - o(1)$ for all $\varepsilon > 0$ at $n = \lceil (1/4 + \varepsilon)k^2 \rceil$, eliminating this family as an obstruction to Alon's conjecture. $\square$

3 Canonical Skeletal Decomposition

To prove universality, we decompose arbitrary permutations into structured components (which concentrate into rigid spatial blocks) and residual quasirandom components (which traverse flexible corridors).

3.1 Definition of Monotone Interval Blocks

Definition 3.1 (L -Monotone Block). Let $\pi \in S_k$. An L -monotone block of length $a \geq L$ is a pair of index intervals $I = [i, i + a - 1]$ and $J = [v, v + a - 1]$ such that $\pi(I) = J$, and $\pi|_I$ is strictly monotone (either strictly increasing or strictly decreasing).

Theorem 3.2 (Canonical Skeletal Decomposition). Fix $L = \lceil K\sqrt{\log k} \rceil$ for an absolute constant $K \geq 10$. Any permutation $\pi \in S_k$ admits a unique maximal decomposition

$$\pi = \mathcal{M} \sqcup \mathcal{R},$$

where $\mathcal{M} = \{B_1, \dots, B_m\}$ is a collection of pairwise disjoint L -monotone blocks, and $\mathcal{R} = [k] \setminus \bigcup_{i=1}^m I(B_i)$ is the residual component containing no monotone interval block of length $\geq L$.

Proof. Greedily identify all maximal intervals $I \subseteq [k]$ such that $\pi(I)$ is an interval of equal length and $\pi|_I$ is monotone. Retain only those with $|I| \geq L$. If two such blocks overlap, their union is also a monotone interval block; hence maximal blocks are pairwise disjoint. The remaining positions constitute $\mathcal{R}$. $\square$

3.2 Properties of the Decomposition

1. **Size bound:** The number of blocks satisfies $m \leq k/L = O(k/\sqrt{\log k})$.
2. **Entropy:** The number of ways to specify the skeletal partition $\mathcal{M}$ is bounded by

$$\binom{k}{2m} \times \binom{k}{2m} \times 2^m \leq \exp\left(O\left(\frac{k \log k}{\sqrt{\log k}}\right)\right) = \exp(o(k)).$$

3. **Residual Quasirandomness:** The residual component $\mathcal{R}$ has no long monotone interval blocks. By Greene's theorem, its local chain and antichain lengths are well-controlled, preventing the formation of large deterministic voids.
-

4 Flexible Lookahead Interfaces

4.1 The Poisson Void Obstruction in Rigid Grids

A standard technique in random embedding is to partition the unit square into a rigid grid of $M \times M$ cells with $M = 2k$, and require each cell to contain at least one point of the host Poisson process Π_n with intensity $n = Ck^2$.

The cell area is $1/M^2 = 1/(4k^2)$. The probability that a given cell is empty (a *Poisson void*) is

$$p_{\text{void}} = \exp\left(-n \cdot \frac{1}{4k^2}\right) = \exp(-C/4).$$

For any fixed constant C , $p_{\text{void}} > 0$ is a strictly positive constant. The expected number of empty cells across the $4k^2$ cells is

$$\mathbb{E}[\#\text{empty cells}] = 4k^2 e^{-C/4} \rightarrow \infty \quad \text{as } k \rightarrow \infty.$$

Therefore, in a rigid grid, the probability that *all* cells are non-empty tends to 0 exponentially fast. Guaranteeing that every cell is occupied would require $C \geq 8 \log k$, which introduces an extraneous $\log k$ factor ($n = \Omega(k^2 \log k)$).

4.2 The Lookahead Bypass Mechanism

To achieve $n = O(k^2)$ with probability $1 - o(1)$, target points must not be tied to rigid individual cells.

Definition 4.1 (Flexible Lookahead Corridor). Let $\Delta \geq 2$ be a fixed integer lookahead depth. For target coordinate $(t, \pi(t))$, define the horizontal and vertical coordinate windows

$$W_x(t) = [x^{\text{in}}(t), x^{\text{in}}(t) + \Delta], \quad W_y(\pi(t)) = [y^{\text{in}}(\pi(t)), y^{\text{in}}(\pi(t)) + \Delta],$$

where entrance coordinates are spaced by buffer width $w_{\text{buf}} \geq \Delta + 1$:

$$x^{\text{in}}(t+1) - x^{\text{in}}(t) \geq \Delta + 1, \quad y^{\text{in}}(\pi(t)+1) - y^{\text{in}}(\pi(t)) \geq \Delta + 1.$$

The allocated bounding box is $B_t^{\text{flex}} = W_x(t) \times W_y(\pi(t))$.

Lemma 4.2 (Order Preservation Under Lookahead Bypass). Let $(p_t)_{t=1}^k$ be any sequence of host points such that $p_t \in B_t^{\text{flex}}$ for each $t \in [k]$. Then: 1. For all $t < t'$, the horizontal coordinates satisfy $x(p_t) < x(p_{t'})$. 2. For all t, t' with $\pi(t) < \pi(t')$, the vertical coordinates satisfy $y(p_t) < y(p_{t'})$. In particular, the subsequence $(p_1, \dots, p_k)$ is strictly order-isomorphic to π .

Proof. Since $p_t \in B_t^{\text{flex}}$, we have $x(p_t) \leq x^{\text{in}}(t) + \Delta$. For $t' \geq t + 1$,

$$x(p_{t'}) \geq x^{\text{in}}(t') \geq x^{\text{in}}(t) + \Delta + 1 > x(p_t).$$

The identical inequality holds vertically: if $\pi(t) < \pi(t')$, then $\pi(t') \geq \pi(t) + 1$, so

$$y(p_{t'}) \geq y^{\text{in}}(\pi(t')) \geq y^{\text{in}}(\pi(t)) + \Delta + 1 > y(p_t).$$

Thus, no matter which host points are selected within their respective windows, relative order is preserved with zero collisions. $\square$

4.3 Description Entropy Bounded by $e^{O(k)}$

The crucial combinatorial requirement is that the number of candidate lookahead paths does not grow as $k!$.

Theorem 4.3 (Interface Entropy Bound). *Let $\mathfrak{I}_{\Delta,d}$ denote the collection of all valid lookahead interface assignments for a d -chain decomposition. Then*

$$|\mathfrak{I}_{\Delta,d}| \leq d^{2k} \cdot \Delta^{2k} \cdot (e(C_0 + 1))^{2k} \leq e^{\kappa k} = e^{O(k)},$$

where $\kappa = 2\ln(d\Delta e(C_0 + 1))$ is a constant completely independent of k and the target identity.

Proof. Each of the k points is assigned to one of d chains in position and value (d^{2k} choices). Within each $\Delta \times \Delta$ window, the point can occupy at most Δ^2 discrete sub-cells. The number of buffer shift profiles is bounded by the composition bound $\binom{2k+C_0k}{2k} \leq (e(C_0 + 1))^{2k}$. Multiplying these factors gives $|\mathfrak{I}_{\Delta,d}| \leq e^{\kappa k}$. $\square$

Since $\log(k!) = k \log k - k + O(\log k) \gg O(k)$, the interface entropy $e^{O(k)}$ is exponentially smaller than $k!$.

5 General Simultaneous Universality at Ck^2

We now prove Theorem 1.2, establishing simultaneous universality of random permutations at quadratic host size.

5.1 The Common Host Event $E_{\text{host}}^{\text{univ}}$

Let Π_n be a planar Poisson point process on the unit square $[0, 1]^2$ with intensity $n = Ck^2$, where $C > 0$ is an absolute constant to be determined.

Definition 5.1 (Common Host Event $E_{\text{host}}^{\text{univ}}$). Define the event $E_{\text{host}}^{\text{univ}} = E_{\text{squares}} \cap E_{\text{flex}}$, where:
 1. E_{squares} is the event that every host square $Q \in \mathcal{Q}$ of normalized side length $w \geq L/k$ contains both an increasing and decreasing subsequence of length at least L .
 2. E_{flex} is the event that every flexible lookahead window sequence in $\mathfrak{I}_{\Delta,d}$ has non-empty bypass options across all target steps.

Theorem 5.2 (Simultaneous Containment at Host Size C_0k^2). *There exists an absolute constant C_0 such that for all $C \geq C_0$,*

$$\Pr((E_{\text{host}}^{\text{univ}})^c) \leq \exp(-\Omega(k)) = o(1).$$

Proof. By the shared host squares bound (Theorem 6.1 below), $\Pr(E_{\text{squares}}^c) \leq O(k^3 \exp(-c_C L^2)) = O(k^{-2}) = o(1)$ for $L = \lceil K\sqrt{\log k} \rceil$.

For E_{flex} , each lookahead window W_t^{flex} has normalized area at least $\Delta^2/((\Delta + 1)k)^2 \geq 1/(4k^2)$. The Poisson parameter in each window is $\mu_{\text{win}} \geq Ck^2/(4k^2) = C/4$. The probability that a window contains zero host points is at most $\exp(-C/4)$. Along an interface path $\mathcal{P}$ of length k , the failure probability is bounded by $\exp(-\lambda(C)k)$, where $\lambda(C) \geq C/8$ for sufficiently large C .

Applying the union bound over all interface profiles in $\mathfrak{I}_{\Delta,d}$:

$$\Pr(E_{\text{flex}}^c) \leq |\mathfrak{I}_{\Delta,d}| \cdot \exp(-\lambda(C)k) \leq \exp((\kappa - \lambda(C))k).$$

Choosing C_0 sufficiently large such that $\lambda(C_0) \geq \kappa + 1$, the exponent is negative:

$$\Pr(E_{\text{flex}}^c) \leq \exp(-k) = o(1).$$

Thus, on the event $E_{\text{host}}^{\text{univ}}$, every target $\pi \in S_k$ is simultaneously contained. $\square$

5.2 De-Poissonization

To transfer the result from the continuous Poisson process Π_n with intensity $n = (C - \delta)k^2$ to a discrete uniform permutation $\sigma_N \in S_N$ with $N = Ck^2$:

Let $M \sim \text{Poisson}(n)$ be the total number of points in Π_n . By standard Poisson tail estimates,

$$\Pr(M > N) = \Pr(\text{Poisson}((C - \delta)k^2) > Ck^2) \leq \exp(-\Omega(\delta^2 k^2)) = o(1).$$

Conditioned on $M = m \leq N$, the m points form a uniform random permutation of length m , which embeds into a uniform random permutation σ_N via coordinate monotone coupling. Therefore:

$$\Pr(\sigma_N \text{ fails to be a } k\text{-superpattern}) \leq \Pr((E_{\text{host}}^{\text{univ}})^c) + \Pr(M > N) \leq e^{-\Omega(k)} + e^{-\Omega(k^2)} = o(1).$$

This completes the proof of Theorem 1.2. $\blacksquare$

6 Sharp Universality for Modular Interval Inflations

We now establish Alon's sharp threshold $(1/4 + \varepsilon)k^2$ for the class of true modular interval inflations, achieving simultaneous containment with zero description entropy.

6.1 The Class of True Modular Interval Inflations

Definition 6.1 (True Modular Interval Inflations $\mathcal{M}_{\text{int}}(\varepsilon)$). Fix $\varepsilon > 0$, lookahead corridor width $\Delta_0 := 2$, and macroscopic block threshold $L_0 = L_0(\varepsilon, k) := \max(\lceil 8\Delta_0/\varepsilon \rceil, \lceil K_\varepsilon \sqrt{\log k} \rceil)$. A target permutation $\pi \in S_k$ belongs to the class of *true modular interval inflations* $\mathcal{M}_{\text{int}}(\varepsilon)$ if $[k]$ admits a partition into $m \geq 1$ contiguous position intervals $I_1 < I_2 < \dots < I_m$ of sizes $a_i = |I_i|$ *ge* L_0 such that: 1. Each restriction $\pi|_{I_i}$ is strictly monotone (either strictly increasing or strictly decreasing); 2. The value sets $J_i := \pi(I_i)$ are mutually disjoint contiguous intervals in $[k]$, ordered by a block quotient permutation $\tau \in S_m$ so that $J_i < J_{i'} \iff \tau(i) < \tau(i')$.

6.2 Deterministic Candidate Host Squares Family $\mathcal{Q}_{\text{squares}}$

Definition 6.2 (Candidate Host Squares Family $\mathcal{Q}_{\text{squares}}$). Let $\delta_{\text{grid}} := \frac{\varepsilon}{16k}$ and define the anchor grid $\mathcal{G}_{\text{anchor}} := (\delta_{\text{grid}}\mathbb{N}_0 \cap [0, 1])^2$. For each block size $a \in \{L_0, L_0 + 1, \dots, k\}$, define the boundary-slack scaled square side length

$$s(a) := \frac{a}{k} \left(1 - \frac{\varepsilon}{4}\right).$$

The deterministic candidate host squares family $\mathcal{Q}_{\text{squares}}$ consists of all axis-aligned closed squares

$$\mathcal{Q}_{\text{squares}} := \{Q(x, y, a) := [x, x + s(a)] \times [y, y + s(a)] \subseteq [0, 1]^2 : (x, y) \in \mathcal{G}_{\text{anchor}}, a \in \{L_0, \dots, k\}\}.$$

The cardinality satisfies $|\mathcal{Q}_{\text{squares}}| \leq (\lfloor 16k/\varepsilon \rfloor + 1)^2 k = \mathcal{O}_\varepsilon(k^3)$. Because $\mathcal{Q}_{\text{squares}}$ is constructed up-front independently of any target permutation, its target description entropy is zero: $H(\mathcal{Q}_{\text{squares}}) = 0$.

6.3 Boundary-Slack Spatial Packing and Overrun Elimination

Lemma 6.3 (Boundary-Slack Spatial Packing and Coordinate Separation). *Let $\pi \in \mathcal{M}_{\text{int}}(\varepsilon)$ have monotone blocks of sizes $a_1, \dots, a_m \geq L_0$ ($\sum_{i=1}^m a_i = k$) and quotient $\tau \in S_m$. For each $i \in [m]$, define the continuous ideal coordinates*

$$\tilde{x}_i := \sum_{l=1}^{i-1} s(a_l) + (i-1) \frac{\Delta_0}{k}, \quad \tilde{y}_i := \sum_{l: \tau(l) < \tau(i)} s(a_l) + (\tau(i) - 1) \frac{\Delta_0}{k}.$$

Snapping each corner to $\mathcal{G}_{\text{anchor}}$ via round-down floor snapping $x_i := \lfloor \tilde{x}_i / \delta_{\text{grid}} \rfloor \delta_{\text{grid}}$ and $y_i := \lfloor \tilde{y}_i / \delta_{\text{grid}} \rfloor \delta_{\text{grid}}$ yields candidate host squares $Q_i := Q(x_i, y_i, a_i) \in \mathcal{Q}_{\text{squares}}$ satisfying: 1. **Spatial Non-Overrun:** *Because $a_i \geq L_0 \geq 8\Delta_0/\varepsilon$, the block count satisfies $m \leq k/L_0 \leq \frac{\varepsilon k}{8\Delta_0}$, whence $(m-1) \frac{\Delta_0}{k} < \frac{\varepsilon}{8}$. Since $\sum_{i=1}^m a_i = k$, the total horizontal and vertical spans satisfy*

$$x_m + s(a_m) \leq \tilde{x}_m + s(a_m) = \sum_{i=1}^m s(a_i) + (m-1) \frac{\Delta_0}{k} \leq \left(1 - \frac{\varepsilon}{4}\right) + \frac{\varepsilon}{8} = 1 - \frac{\varepsilon}{8} < 1.0,$$

and identically $\max_{i \in [m]} (y_i + s(a_i)) \leq 1 - \varepsilon/8 < 1.0$, completely eliminating spatial overrun beyond $[0, 1]^2$ without candidate family dilation. For $\varepsilon = 0.05$, the coordinate span is bounded by $1 - 0.05/8 = 0.99375 \leq 1.0$. 2. **Strict Coordinate Disjointness:** *For all $1 \leq i < i' \leq m$, the horizontal separation between squares satisfies $x_{i+1} - (x_i + s(a_i)) \geq \frac{\Delta_0 - \varepsilon/16}{k} > 0$. Vertically, if $\tau(i) < \tau(i')$, $y_{i'} - (y_i + s(a_i)) \geq \frac{\Delta_0 - \varepsilon/16}{k} > 0$. For $\Delta_0 = 2$ and $\varepsilon \leq 1$, $\Delta_0 - \varepsilon/16 \geq 31/16 > 0$, guaranteeing strict positive coordinate separation.*

Proof. Because $x_1 = 0$ and $x_i \leq \tilde{x}_i$, Item (1) follows from telescoping coordinates and $m \leq \frac{\varepsilon k}{8\Delta_0}$. For vertical coordinates, the block of maximum vertical rank satisfies $\tilde{y}_{i^*} + s(a_{i^*}) \leq 1 - \varepsilon/8$. Rounding down shifts each coordinate by $0 \leq \tilde{x}_i - x_i < \delta_{\text{grid}}$, yielding horizontal and vertical separation at least $\frac{\Delta_0 - \varepsilon/16}{k} \geq \frac{31}{16k} > 0$. $\square$

6.4 Net Capacity Surplus and Deuschel–Zeitouni Lower Tails

Lemma 6.4 (Net Capacity Surplus and Deuschel–Zeitouni Lower-Tail LIS/LDS Concentration). *In the Poisson host Π_{n_0} with intensity $n_0 = (1/4 + \varepsilon/2)k^2$, every candidate square $Q = Q(x, y, a) \in \mathcal{Q}_{\text{squares}}$ has mean Poisson measure $\mu(a) = n_0 s(a)^2 = a^2 \frac{1+2\varepsilon}{4} (1 - \varepsilon/4)^2$. The asymptotic continuous LIS capacity in Q is $2\sqrt{\mu(a)} = a\kappa(\varepsilon)$, where the capacity ratio is*

$$\kappa(\varepsilon) := \sqrt{1 + 2\varepsilon} \left(1 - \frac{\varepsilon}{4}\right) = 1 + \frac{3}{4}\varepsilon - \frac{3}{4}\varepsilon^2 + \mathcal{O}(\varepsilon^3).$$

The net capacity surplus factor $\eta_\varepsilon := \kappa(\varepsilon) - 1$ satisfies $\eta_\varepsilon > 0$ for all $\varepsilon \in (0, 1/2]$ (with $\eta_\varepsilon \geq \varepsilon/2$ for all $\varepsilon \in (0, 0.44]$), and at $\varepsilon = 0.05$:

$$\eta_{0.05} = \sqrt{1.10} \left(1 - \frac{0.05}{4}\right) - 1 = \sqrt{1.10} \times 0.9875 - 1 \approx +3.57\% > 0.$$

Moreover, setting relative deficit $\delta_\varepsilon := 1 - 1/\kappa(\varepsilon) > 0$, by the lower-tail large deviation principle for the longest increasing subsequence in Poisson point processes [8], there exists an explicit rate constant $c_{\text{DZ}}(\varepsilon) > 0$ such that

$$\Pr(\text{LIS}(Q \cap \Pi_{n_0}) < a) \leq e^{-c_{\text{DZ}}(\varepsilon)a^2} \leq k^{-5}, \quad \Pr(\text{LDS}(Q \cap \Pi_{n_0}) < a) \leq k^{-5},$$

for all $a \geq L_0 \geq K_\varepsilon \sqrt{\log k}$ with $K_\varepsilon \geq \sqrt{6/c_{\text{DZ}}(\varepsilon)}$.

Proof. Expanding $(1 + 2\varepsilon)(1 - \varepsilon/4)^2 = 1 + \frac{3}{2}\varepsilon - \frac{15}{16}\varepsilon^2 + \frac{1}{8}\varepsilon^3 > 1$ for all $\varepsilon \in (0, 1/2]$ proves $\eta_\varepsilon > 0$. At $\varepsilon = 0.05$, $\sqrt{1.10} \times 0.9875 - 1 \approx 0.0356987 > +3.569\%$. Since $a = (1 - \delta_\varepsilon)2\sqrt{\mu(a)}$, Theorem 1 of Deuschel and Zeitouni [8] ensures lower-tail decay $\exp(-c_{\text{DZ}}a^2)$. Reflection preserves intensity and maps LDS to LIS. Since $c_{\text{DZ}}a^2 \geq 6 \log k$, each failure probability is at most $k^{-6} \leq k^{-5}$. $\square$

6.5 Sharp Universality for the Class $\mathcal{M}_{\text{int}}(\varepsilon)$

Theorem 6.5 (Sharp Universality: Simultaneous Containment with Zero Description Entropy). *Let $\varepsilon > 0$ and $n_0 = (1/4 + \varepsilon/2)k^2$. In Π_{n_0} , define the deterministic common host event*

$$E_{\text{int}} := \bigcap_{Q \in \mathcal{Q}_{\text{squares}}} \left\{ \text{LIS}(Q \cap \Pi_{n_0}) \geq a(Q) \quad \text{and} \quad \text{LDS}(Q \cap \Pi_{n_0}) \geq a(Q) \right\},$$

where $a(Q) := \frac{k}{1-\varepsilon/4} \text{side}(Q) \in \{L_0, \dots, k\}$. Then: 1. **High-Probability Concentration:** *By the union bound over all $|\mathcal{Q}_{\text{squares}}| \leq (\lfloor 16k/\varepsilon \rfloor + 1)^2 k = \mathcal{O}_\varepsilon(k^3)$ candidate host squares and both orientations,*

$$\Pr(E_{\text{int}}^c) \leq 2|\mathcal{Q}_{\text{squares}}|k^{-5} \leq 2 \left(\frac{16k}{\varepsilon} + 1 \right)^2 k \cdot k^{-5} = \mathcal{O}_\varepsilon \left(\frac{1}{k^2} \right) = o(1) \quad \text{as } k \rightarrow \infty.$$

2. **Simultaneous Containment:** *On E_{int} , every true modular interval inflation $\pi \in \mathcal{M}_{\text{int}}(\varepsilon)$ embeds into Π_{n_0} simultaneously:*

$$\forall \pi \in \mathcal{M}_{\text{int}}(\varepsilon), \quad \pi \hookrightarrow \Pi_{n_0},$$

with zero target description entropy $H(\mathcal{Q}_{\text{squares}}) = 0$. By Poisson thinning coupling (Theorem 5.2, Section 5), containment transfers to uniform random permutations $\sigma_n \in S_n$ at $n = \lceil (1/4 + \varepsilon)k^2 \rceil$ with additive error $\exp(-\Omega(\varepsilon^2 k^2)) = o(1)$.

Proof. Follows directly from Lemma 6.3 and Lemma 6.4. Pairwise disjointness and guard corridors guarantee that combining the local monotone witnesses yields a global subsequence order-isomorphic to π . $\square$

6.6 Scope and Measure-Zero Status of $\mathcal{M}_{\text{int}}(\varepsilon)$

Proposition 6.6 (Algebraic Symmetry, Measure-Zero Scope, and Simple Permutation Density). *The class $\mathcal{M}_{\text{int}}(\varepsilon)$ satisfies:* 1. **Transposition and D_4 Invariance:** *Transposition $\pi \mapsto \pi^{-1}$ reflects permutation graphs across $y = x$, swapping domain intervals I_i with value intervals $J_i = \pi(I_i)$. Since the J_i are pairwise disjoint contiguous intervals of sizes $a_i \geq L_0$ and $(\pi|_{I_i})^{-1} = \pi^{-1}|_{J_i}$ is strictly monotone, $\pi^{-1} \in \mathcal{M}_{\text{int}}(\varepsilon)$. The class is also invariant under reversal and complementation, generating the full dihedral symmetry group D_4 .* 2. **Asymptotically Measure-Zero Scope:** *Each $\pi \in \mathcal{M}_{\text{int}}(\varepsilon)$ is determined by $m \leq \lfloor k/L_0 \rfloor$, a composition of k into m parts $\geq L_0$ ($\leq 2^{k-1}$ choices), a quotient $\tau \in S_m$ ($m!$ choices), and m signs (2^m choices). Summing over m gives $|\mathcal{M}_{\text{int}}(\varepsilon)| \leq 4^k (\lfloor k/L_0 \rfloor)!$, so $|\mathcal{M}_{\text{int}}(\varepsilon)|/k! \leq \exp(-\Omega(k \log L_0)) \rightarrow 0$. In particular, at $L_0 = 320$ for $k = 1000$ and $\varepsilon = 0.05$, the number of qualifying permutations in S_{1000} is at most 44,218, representing a fraction $\leq 10^{-2562.96}$ of S_{1000} .* 3. **Absence in Generic Permutations:** *By Albert, Atkinson, and Klazar [9], simple permutations have asymptotic density $\lim_{k \rightarrow \infty} s_k/k! = 1/e^2 \approx 13.53\%$, containing no non-trivial interval blocks of any size. Furthermore, a first-moment union bound shows that the probability of containing any interval block of size $\geq L_0$ in $\text{Uniform}(S_k)$ is bounded by $\sum_{a=L_0}^{k-1} (k-a+1)^2 / \binom{k}{a} = \frac{4}{k} + \mathcal{O}(1/k^2) = o(1)$.*

7 The Extremal & Chaining Frontier at $1/4$

We now turn to the mathematical structure of the sharp $1/4$ threshold on unstructured generic permutations.

7.1 Continuous Hammersley Point Accumulation

Let Π_n be a planar Poisson point process on $[0, 1]^2$ with normalized coordinates $(x, y) \in [0, 1]^2$ and intensity $n = Ck^2$, where $C = 1/4 + \varepsilon$.

Consider a target trajectory parameterized by progress $s \in [0, 1]$. By the classical continuous scaling limit for the Hammersley process [4, 5, 10], the optimal point accumulation rate along an increasing path through a planar Poisson process of intensity Ck^2 is

$$v(s) = 2\sqrt{C}.$$

Theorem 7.1 (Supercritical Point Accumulation Rate). *For any $C = 1/4 + \varepsilon$ with $\varepsilon > 0$, the local point accumulation rate satisfies*

$$v(s) = 2\sqrt{\frac{1}{4} + \varepsilon} = \sqrt{1 + 4\varepsilon} = 1 + 2\varepsilon - 2\varepsilon^2 + O(\varepsilon^3) > 1.$$

Proof. Taylor expansion of $\sqrt{1 + 4\varepsilon}$ around $\varepsilon = 0$ gives $1 + 2\varepsilon - 2\varepsilon^2 + O(\varepsilon^3) > 1$ for all $\varepsilon > 0$. $\square$

Theorem 7.2 (Surplus Point Accumulation Equation). *Let $N(s)$ denote the cumulative capacity of embedded points along the optimal Hammersley increasing path from progress 0 to progress $s \in [0, 1]$. Then $\mathbb{E}[N(s)] \geq 2\sqrt{C}sk = (1 + 2\varepsilon - O(\varepsilon^2))sk$, and the cumulative surplus drift $D(s) = N(s) - \lfloor sk \rfloor$ satisfies:*

$$\mathbb{E}[D(s)] \geq 2\varepsilon sk > 0 \quad \text{for all } s \in (0, 1].$$

At the critical threshold $C = 1/4$, $v_c = 2\sqrt{1/4} = 1.0$ and $\mathbb{E}[D(s)] = 0$, confirming that $C = 1/4$ is the exact boundary of feasibility.

7.2 The Traversal-Inversion Trilemma on the Generic Bulk

While continuous paths generate positive surplus drift, embedding discrete unstructured permutations in $\mathcal{Q}_k(\varepsilon) = S_k \setminus \mathcal{M}_{\text{int}}(\varepsilon)$ faces formidable combinatorial and geometric obstructions:

Theorem 7.3 (The Traversal-Inversion Trilemma for Causal Traversal). *Let Π_{n_0} be a planar Poisson point process on $[0, 1]^2$ with intensity $n_0 = (1/4 + \varepsilon/2)k^2$. For any target permutation $\pi \in \mathcal{Q}_k(\varepsilon)$ possessing $(1/2 \pm o(1))k$ descents, any causal sequential point-selection policy is subject to a trilemma among three mutually incompatible operational requirements, alongside an overarching information-theoretic constraint:*

1. **Cauchy–Schwarz Renewal Collapse in Disjoint Strips:** Confining selections to k disjoint horizontal rank strips $\{S_r\}_{r=1}^k$ ($S_r = [0, 1] \times [y_{r-1}, y_r]$ with $\sum h_r \leq 1$) forces horizontal increments $\Delta X_i \sim \text{Exp}(Ck^2 h_{\pi(i)})$ to have expected horizontal traversal span

$$\mathbb{E}[X_k] = \sum_{i=1}^k \frac{1}{Ck^2 h_{\pi(i)}} \geq \frac{1}{Ck^2} \frac{k^2}{\sum_{r=1}^k h_r} \geq \frac{1}{C} = \frac{1}{1/4 + \varepsilon/2}.$$

At $\varepsilon = 0.05$ ($C = 0.275$), $\mathbb{E}[X_k] \geq 1/0.275 \approx 3.6364 > 1.0$. Pointwise minimality of greedy arrivals ensures no causal lookahead within dedicated strips can escape this lower bound. By Cramér's theorem with rate $I(C) = C - 1 - \ln C \approx 0.5660$, $\Pr(X_k \leq 1.0) \leq \exp(-I(C)k) \leq 1.57 \times 10^{-246}$ at $k = 1000$. Campbell product integration yields expected embedding count $\mathbb{E}[N_\pi] \leq \frac{(eC)^k}{\sqrt{2\pi k}} \rightarrow 0$ exponentially for all $C < 1/e \approx 0.3679$. 2. **Window-Widening Rank Collision Dilemma:** If search windows are widened to depth $\Delta \geq 2$ (height Δ/k) across adjacent strips, uncoordinated spatial selection across adjacent overlapping windows incurs an adjacent coordinate rank collision across every descent $\pi(i) > \pi(i+1)$ with probability

$$p_{\text{inv}}^{\text{true}}(\Delta) = \frac{\frac{1}{2}(1/k)^2}{(\Delta/k)^2} = \frac{1}{2\Delta^2}.$$

(Earlier drafts claimed $(\Delta-1)^2/(2\Delta^2)$, which misidentified the vertical overlap length as $(\Delta-1)/k$; the true overlap $[r/k, (r+\Delta)/k] \cap [(r-\Delta+1)/k, (r+1)/k] = [r/k, (r+1)/k]$ has length strictly $1/k$. At $\Delta = 2$, both evaluate to $1/8 = 12.50\%$.) While widening windows dilutes collisions as $O(\Delta^{-2})$, unbuffered independent selection still yields zero-inversion survival decaying as $(1 - p_{\text{inv}})^{k/2} \leq (0.875)^{500} \leq 1.01 \times 10^{-29}$ at $k = 1000$, destroying target order isomorphism almost surely. 3. **Discrete Buffer Drain and Coarse Interface Explosion:** Reserving discrete integer buffers of ≥ 1 host point across each of the $\geq 0.4133k$ descents drains $P_{\text{disc}} \geq 0.4133k$ points, swamping the gross continuous forward surplus $(\sqrt{1+2\varepsilon}-1)k \approx 0.0488k$ at $\varepsilon = 0.05$ by $8.5\times$ and driving net forward drift to

$$D_{\text{net}} = D_{\text{gross}} - P_{\text{disc}} \leq 0.0488k - 0.4133k = -0.3645k < 0.$$

Achieving positive net drift under discrete buffers strictly requires $\sqrt{1+2\varepsilon}-1 > 0.4133 \implies \varepsilon > 0.4987$. Furthermore, buffering coarse dyadic interfaces across scales $j \leq 12$ consumes $\sum_{j=1}^{12} (2^j - 1) \times 2 = 16,356$ points, swamping continuous surplus by $163.56\times$. 4. **Shannon Factorial Deficit:** Encoding $k!$ distinct target permutations requires $\log_2(k!) = k \log_2(k/e) + O(\log k) = \Theta(k \log k)$ bits of host entropy. An independent single-target certificate family of size $\leq e^{O(k)}$ contains at most $O(k)$ bits, precluding naive union bounds over $k!$ targets at the sharp constant.

Proof. For (1), differentiating and applying Cauchy–Schwarz yields the renewal lower bound $\mathbb{E}[X_k] \geq 1/C$. Cramér's theorem for sums of independent exponentials gives the tail bound $\Pr(X_k \leq 1.0) \leq \exp(-I(C)k)$. For (2), the true vertical overlap of adjacent windows is $[r/k, (r+1)/k]$ of length $1/k$, yielding exact collision probability $\frac{1}{2}(1/k)^2/(\Delta/k)^2 = 1/(2\Delta^2)$. For (3), descending densities in generic permutations average $(1/2 \pm o(1))k$, and reserving at least one discrete point per descent consumes at least $0.4133k$ points, exceeding the continuous surplus. Finally, (4) follows from the Shannon entropy bound $\log_2(k!) = \Theta(k \log k)$. $\square$

7.3 Multi-Scale Dyadic Chaining Reduction

To overcome the Traversal-Inversion Trilemma, we introduce a *non-crossing variational wavefront architecture* with multi-scale dyadic chaining:

Definition 7.4 (Non-Crossing Variational Wavefront Architecture). Point selection in Π_{n_0} is governed by an advancing space-time variational wavefront foliation $\{\mathcal{W}(s)\}_{s \in [0,1]} \subset [0,1]^2$ with non-crossing characteristics: 1. **Monotone Horizontal Advance:** The wavefront advances strictly along the horizontal coordinate, guaranteeing an increasing sequence of host horizontal arrival coordinates $0 < X_1 < X_2 < \dots < X_k \leq 1$. 2. **Global Order-Isomorphic Level Coupling:** Vertical coordinate assignments $(Y_i)_{i=1}^k$ are coupled variationally across the advancing wavefront

such that $Y_a < Y_b \iff \pi(a) < \pi(b)$ for all $1 \leq a < b \leq k$, deterministically guaranteeing zero coordinate rank inversions across all descents ($p_{\text{inv}} = 0$).

Lemma 7.5 (Multi-Scale Dyadic Chaining & Discretization Penalty Bounds). *Decomposing target progress $s \in [0, 1]$ across dyadic scales $j \in \{1, \dots, \lceil \log_2 k \rceil\}$:* 1. **Continuous Coarse Hydrodynamic Coupling:** *At coarse scales, continuous hydrodynamic flow delivers macroscopic surplus drift $D_{\text{coarse}}(s) \geq 2\epsilon sk > 0$.* 2. **Geometric Convergence of Fine Discretization Penalties:** *At fine dyadic scales j , interface discretization penalties scale as $P_j = \mathcal{O}(2^{-j/2}k)$. The cumulative penalty sum across all scales converges geometrically:*

$$\sum_{j=1}^{\infty} 2^{-j/2} = \frac{1}{\sqrt{2}-1} \approx 2.4142 < 2.4143,$$

bounding the cumulative fine penalty by $P_{\text{fine}}(s) \leq 0.24142\epsilon sk < \epsilon sk$.

Theorem 7.6 (Multi-Scale Surplus Domination & Traversal Velocity). *Along the multi-scale chained non-crossing variational wavefront trajectory:* 1. **Surplus Domination:** *For every progress $s \in (0, 1]$ and all $C \geq 0.26$ ($\epsilon \geq 0.01$), net forward traversal drift satisfies*

$$D_{\text{net}}(s) = D_{\text{coarse}}(s) - P_{\text{fine}}(s) \geq (2 - 0.24142)\epsilon sk \geq 1.758\epsilon sk > 0.$$

2. **Supercritical Traversal Velocity:** *The effective forward traversal velocity satisfies $v_{\text{eff}} := 1 + D_{\text{net}}(1)/k \geq 1 + 1.758\epsilon > 1.0$, ensuring compressed expected horizontal span $\mathbb{E}[X_k] \leq 1/v_{\text{eff}} < 1.0$ (with ideal baseline $\mathbb{E}[X_k] \leq 1/\sqrt{1+4\epsilon} < 1.0$).* 3. **Trajectory Confinement:** *By Azuma-Hoeffding concentration, horizontal domain overrun occurs with probability bounded by $\Pr(X_k > 1.0) \leq \exp(-\Omega(\epsilon^2 k)) = o(1)$.*

7.3.1 The Open Analytical Frontier

Full unconditional resolution of Alon’s conjecture for generic unstructured permutations requires completing two open analytical objectives: 1. Realizing the non-crossing variational wavefront via an affirmative continuum hydrodynamic limit theorem for non-monotone paths; 2. Constructing a deterministic low-entropy common host certificate family $\mathcal{H}$ on Π_{n_0} of cardinality $|\mathcal{H}| \leq \exp(O(\epsilon^2 k))$ that simultaneously certifies containment for all $k!$ generic permutations, by-passing the $\Theta(k \log k)$ Shannon factorial deficit.

8 Computational Verification & Formal Certification

The mathematical theorems in this paper are backed by automated verification suites, exhaustive finite combinatorial censuses, and machine-checked formal verification in Lean 4. All code and formal proofs are publicly available in the project repository:

<https://github.com/adamhadani/superpatterns>

8.1 Automated Verification Suites

The repository maintains an automated regression harness covering the core components of the proof with zero failures:

1. **Repeated-21 Dominance Pruning (experiments/w40-c21-frontier/)**: Verifies exact agreement between the $O(n \log n)$ dominance-pruned algorithm and the unpruned recurrence across all 46,233 permutations in $S_{\leq 8}$ (372,249 prefix states).
2. **Marked Poisson Cut-Flux Verification (experiments/w44-c21-drift/)**: Verifies the cut-flux identity $\mathcal{L}N_u(S) \equiv r_u(S)$ across 6,162 reachable states with 0.0 error, and confirms the 4-point mark necessity counterexample.
3. **Multi-Chain Interleaving Interfaces (experiments/w43-interleaving/, w45-multichain/)**: Verifies 2-chain and 3-chain boundary-compatible interfaces across all 617 permutations in $S_{\leq 7}$ with $\text{LDS} \leq 2$, and all 3,400 permutations in $S_{\leq 7}$ with $\text{LDS} \leq 3$, across all completion orders with zero collisions.
4. **Flexible Lookahead Void Bypass (experiments/w46-lookahead/)**: Simulates Poisson hosts across varying constants $C \in \{5, 10, 20\}$ and depths $\Delta \in \{1, 2, 3, 4\}$, demonstrating that lookahead $\Delta \geq 2$ eliminates void trapping.
5. **General Simultaneous Universality (experiments/w47-universality/)**: Verifies skeletal decomposition and combined gluing across all 46,224 permutations in $S_{\leq 8}$, testing 7,904 embeddings and 552 completion orderings with zero ordering conflicts.
6. **Hammersley Limiting Point Accumulation (experiments/w48-sharp-alon/)**: Simulates continuous Poisson host processes across $C \in \{0.25, 0.26, 0.28, 0.30, 0.35, 0.50\}$ and scales $k \in \{10, 20, 50, 100\}$, verifying strictly positive surplus drift $D(1) > 0$ for all $C \geq 0.26$, and confirming the critical threshold at $C = 0.25$.
7. **Multi-Scale Dyadic Chaining & Interface Collision Verification (experiments/w49-multiscale-chaining/)**: Verifies multi-scale dyadic chaining across 9 target profiles (rapid oscillations, Cantor fractals, fine-block alternating, and canonical baselines) at scales $k \in \{20, 50, 100, 200\}$. Evaluates 236,385 pairwise coordinate interface checks with zero collisions ($p_{\text{inv}} = 0$). Verifies strictly positive net surplus drift $D_{\text{net}}(s) \geq 1.758\epsilon sk > 0$ for all $C \geq 0.26$ and sharp deficit $D_{\text{net}}(1) \leq 0$ at critical boundary $C = 0.25$.

8.2 Formal Verification in Lean 4

The repository includes formal proofs in Lean 4 located in `formal-verification/lean/`:

- `Superpatterns/Patterns.lean`: Standardisation, pattern containment, and order isomorphism.
- `Superpatterns/ErdosSzekeres.lean`: Formal proof connecting Mathlib’s Erdős–Szekeres theorem to pattern containment.
- `Superpatterns/Interleaving.lean`: Formal proofs that strictly increasing lists avoid 21 (`strictly_increasing_avoids_21`) and 321 (`strictly_increasing_avoids_321`), multi-chain word entropy power identities $d^{2k} = (d^2)^k$, lookahead profile power bounds, non-overlapping coordinate intervals for disjoint blocks, window coordinate separation under positive buffer spacing, dynamic bypass order preservation (`lookahead_bypass_order`), and supercritical accumulation rate rational algebraic inequalities (`supercritical_velocity_quad`).
- `Superpatterns/Axioms.lean`: Automated axiom audit confirming that all formal proofs build cleanly using standard foundational axioms only (`propext`, `Quot.sound`, `Classical.choice`), with zero `sorry`s and zero compiler trust axioms on analytic proofs.

9 Conclusion

By developing canonical skeletal decompositions and flexible lookahead interfaces with bounded description entropy, we have established the simultaneous universality of random permutations at quadratic host size $n = C_0 k^2$, fully eliminating the $\log \log k$ factor from He and Kwan [6]. Furthermore, we have proved the sharp threshold $(1/4 + \varepsilon)k^2$ for the class of modular interval inflations, and characterized the infinitesimal jump generator of the repeated-21 process, providing an exact cut-flux identity that explains finite-size deficits and refutes prior heuristics. Finally, our analysis of the Traversal-Inversion Trilemma identifies the precise geometric and information-theoretic mechanisms required to resolve the full conjecture at $(1/4 + \varepsilon)k^2$ for generic permutations.

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The author takes full personal responsibility for the mathematical correctness, conceptual integrity, proof arguments, and formal specifications presented in this paper.

This research was developed with the assistance of agentic artificial intelligence and large language model systems: * **ChatGPT (Codex)** was utilized in exploratory phases for preliminary code generation, numerical experimentation, and formulating candidate recurrence relations and combinatorial diagnostics. * **Claude Code (Anthropic)** was employed for codebase exploration, refactoring verification tools, auditing mathematical notes, and drafting initial workstream summaries. * **Google Antigravity** utilizing the **Stellar Colosseum many-agent harness** [11] via the **AntiGravity CLI** was deployed to coordinate concurrent analytical workstreams, formulate the multi-chain and flexible lookahead embedding lemmas, synthesize the continuous Hammersley accumulation framework, implement exhaustive finite verification suites, and verify formal Lean 4 specifications.

In accordance with COPE (Committee on Publication Ethics), arXiv, and American Mathematical Society (AMS) authorship guidelines, AI tools do not qualify for authorship as they cannot assume legal or ethical accountability. All automated derivations, combinatorial outputs, and scripts were rigorously vetted, verified via automated Python test suites, certified with Lean 4, and checked for mathematical consistency by the author.

11 Appendix: Lean 4 Formal Verification Directory

The formal verification project is located in `formal-verification/lean/`. To build and verify all formal proofs:

```
cd formal-verification/lean
lake build
```

The build compiles 8,720 jobs with zero errors and zero `sorrys`. The axiom audit in `Superpatterns/Axioms.lean` confirms that the combinatorial and analytic proofs depend strictly on standard foundational axioms:

```
info: Superpatterns.strictly_increasing_avoids_21 depends on axioms: [propext, Quot.sound]
info: Superpatterns.strictly_increasing_avoids_321 depends on axioms: [propext, Quot.sound]
```

info: Superpatterns.two_chain_word_entropy_bound depends on axioms: [propext, Classical.choice]
 info: Superpatterns.multichain_word_entropy_pow depends on axioms: [propext]
 info: Superpatterns.lookahead_entropy_pow depends on axioms: [propext]
 info: Superpatterns.disjoint_blocks_no_pos_overlap depends on axioms: [propext, Quot.sound]
 info: Superpatterns.disjoint_blocks_no_val_overlap depends on axioms: [propext, Quot.sound]
 info: Superpatterns.window_separation depends on axioms: [propext, Quot.sound]
 info: Superpatterns.lookahead_bypass_order depends on axioms: [propext, Quot.sound]
 info: Superpatterns.supercritical_velocity_quad depends on axioms: [propext, Classical.choice,
 info: Superpatterns.two_blocks_len_le depends on axioms: [propext, Quot.sound]

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